Huan Wang

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Profile

Dr. Huan Wang is an Engineer/Researcher at the Norwegian Geotechnical Institute (NGI) with more than 10 years of experience in offshore geotechnical engineering. His expertise spans fundamental soil mechanics, soil-structure interaction, physical and numerical modelling, and the design of foundations and anchors for offshore energy systems. He holds a Ph.D. in Geotechnical Engineering from Zhejiang University and has conducted research at Delft University of Technology, the University of Western Australia, and the Hong Kong University of Science and Technology. Having worked across Asia, Australia, Europe, and North America, he has developed a strong international track record. His research integrates advanced physical modelling, including element testing, reduced-scale field testing, and enhanced-gravity centrifuge experiments, with numerical modelling to improve fundamental understanding and develop practical, design-oriented tools. Currently based at NGI's U.S. office, Dr. Wang applies this research background directly to complex offshore engineering projects, bridging fundamental research and engineering practice. He has contributed to major international research and industry programmes such as MIDAS, GEOLAB, PIGS, and TAILWIND, has published extensively in leading journals, and actively collaborates with industry and academic partners worldwide.

Research interests

- Soil mechanics and modelling
- Physical and numerical modelling
- Soil-structure interaction problems
- Offshore foundation/anchor design
- Site investigation

Education

Zhejiang University, Hangzhou, China

Ph.D., Geotechnical engineering, 2020

Supervisor: **Prof. Lizhong Wang**

Dissertation: Experimental and numerical investigations into the behaviour of monopile and suction bucket foundation in sand for offshore wind turbine

Tongji University, Shanghai, China

B.Sc., Civil engineering, 2014

Supervisor: **Prof. Zhiyi Chen**

Dissertation: Seismic Design of Immersed Tunnel Considering Travel Wave Effect

Work experience

Norwegian Geotechnical Institute, Boston, USA

Engineer/Researcher, 2024-present, Boston office

- Geotechnical research and consultancy for offshore energy production system

Norwegian Geotechnical Institute, Oslo, Norway

Engineer/Researcher, 2022-2024, Computational geomechanics, Department of Offshore Energy

- Geotechnical research and consultancy for offshore energy production system

Delft University of Technology, Delft, Netherlands

Post-doc researcher, 2020-2022

Supervisor: **Prof. Amin Askarinejad & Prof. Federico Pisanò & Prof. Ken Gavin**

- Monopile Improved Design through Advanced cyclic Soil modelling (**MIDAS**), *supported by the Netherlands Enterprise Agency (RVO) and TKI Wind op Zee*

  The MIDAS' research programme is led by TU Delft in cooperation with Deltares, and benefits from the interaction with six industry partners along the whole offshore wind supply chain, including wind farm developers (Shell, Eneco, TotalEnergies and RWE), wind turbine developers (Siemens Gamesa Renewable Energy), marine contractors (Huisman, IHC and Van Oord), and research institutes (TU Delft and Deltares). Additionally, the Norwegian Geotechnical Institute (NGI) and DNV are the members of an external technical board. The project aims to improve the understanding on the long-term cyclic response of monopile foundations in sand through advanced physical and numerical modelling, and to develop novel engineering models/tools for monopile design.

- **GEOLAB**: innovative solutions for critical infrastructure, *funded by the European Union **H2020** Research and Innovation Programme*

  The GEOLAB Research Infrastructure (RI) consists of 11 unique installations in Europe aimed to study subsurface behaviour and the interaction with structural CI elements and the environment. The overarching aim of GEOLAB is to integrate and advance these key national research infrastructures towards a one-stop-shop of excellent physical research infrastructure for performing ground-breaking research and innovation to address challenges faced by the Critical Infrastructure of Europe.

Hong Kong University of Science and Technology, Hong Kong, China

Visiting scholar, 2017-2018

Supervisor: **Prof. Charles W W Ng**

Activity: Centrifuge modelling of monopile foundations in sand and clay (HDEC-ZJU-HKUST)

PowerChina Huadong Engineering Corporation Limited (HDEC), the largest offshore wind design enterprise in China, commissioned Zhejiang University (ZJU) and the Hong Kong University of Science and Technology (HKUST) to conduct a collaborative research programme. The study aimed to systematically investigate the monotonic and cyclic lateral response of monopile foundations in sand and clay using an extensive centrifuge modelling campaign.

University of Western Australia, Perth, Australia

Visiting scholar, 2018-2020

Supervisor: **Prof. Barry Lehane & Prof. Fraser Bransby**

Activity: Design methods to quantify lateral and rotational stiffness of short monopiles (UWA-ZJU)

In this joint research project between Zhejiang University (ZJU) and the University of Western Australia (UWA), a series of medium-scale field tests were conducted to quantify the contribution of pile base resistance to the lateral stiffness and capacity of rigid piles subjected to lateral loading in sand. Based on the experimental findings, a CPT-based approach was also developed to evaluate the geotechnical lateral bearing capacity of rigid piles in sand.

Professional service & Professional memberships

- 2022-present: Associate Editor, **International Journal of Physical Modelling in Geotechnics**
- 2025-present: Young Editorial Board Member, **Journal of Zhejiang University-SCIENCE A**
- 2022-present: Member, American Society of Civil Engineers (ASCE)
- 2022-present: Committee member, Deep Foundations Committee of the Geo-InstituteASCE
- 2022-present: Committee member, Soil Properties and Modeling (SP&M) Committee of the Geo-InstituteASCE

Honors and Awards

- **Bright Spark Lecture**, 11th International Conference on Physical Modelling in Geotechnics (ICPMG2026)

- **Best Cyclic Test Predicition & 2nd Best Overall Prediction**, GEOLAB Blind Prediction Contest on Piles under Monotonic and Cyclic Loading
- **OSIGp Prize 2023** (finalist), Society for Underwater Technology, Offshore Site Investigation and Geotechnics Perth

Research project

- 2025-2027: Core research team member of the JIP "**Cost Optimized Foundation Design (COFD): Estimating CPT And Soil Parameters By 3GEO**" led by NGI and ZJU. Funded by CNOOC and QIMG, – Value: 1.2 M¥
- 2024-2027: Core research team member of the JIP "**Monopile Improved Design through Advanced cyclic Soil modelling in clay (MIDASclay)**" led by TU Delft. Funded by RVO (Dutch Enterprise Agency) + industry co-funding– Value: 1.75 M€; Industry partners: Huisman, RWE, Siemens Gamesa, TotalEnergies, Van Oord, Vattenfall; Research partners: Deltares, NGI.
- 2023-2027: Technical board member of the EU project "**TAILWIND- Sustainable station-keeping systems for floating wind**" led by NGI. Funded by EU research council– Value: 6 M€; Project partners: TU Delft, DTU, SINTEF Ocean, Fundación Tecnalia Research & Innovation, Nautilus Floating Solutions, Bekaert Wire Rope Industry, Subsea 7, Fondazione ICONS, Clarke Modet, NKT, Bridon, University of Southampton.
- 2023-2027: Core research team member of the JIP "**PIGS: Piling in Glauconitic Sand**" led by NGI. Funded by industry partners – Value: 5 M$ (Phases 1-4, Phase 4 currently being set up); Industry partners: Ørsted, Equinor, RWE, Avangrid, Attentive Energy; Research partners: University of Massachusetts at Amherst, University of Massachusetts at Dartmouth, Rutgers University, University of Arkansas.
- 2024-2026: Project manager and principle investigator for the industry research project "**Lateral response of piles in sands**" led by NGI. Funded by BP – Value: 140 K$
- 2024-2032: Team member of "**ProWind-Norwegian Infrastructure Platform for Foundation Technology Research in Offshore Wind**" led by NGI. Funded by Research Council of Norway– Value: 89 MNOK; Project partners: NTNU, OsloMet, NIVA, Akvaplan-niva, and the Marine Energy Test Centre.
- 2020-2024: Core research team member of the JIP "**MIDAS: Monopile Improved Design through Advanced Soil cyclic modelling**" at TU Delft. Funded by RVO (Dutch Enterprise Agency) + industry co-funding– Value: 1.3 M€; Industry partners: Eneco, Huisman, IHC, RWE, Shell, Siemens Gamesa, TotalEnergies, Van Oord, Vattenfall; Research partners: Deltares, NGI, DNV.
- 2022-2025: Core research team member of the EU project "**GEOLAB**" at TU Delft. Funded by the European Union H2020 Research and Innovation Programme – Value: 5 M€; partners: Delatres, TU Delft, ETH Zürich, University of Cambridge, NGI, Centro De Estudios Y Experimentacion De Obras Publicas, Gustave Eiffel University, Technical University of Darmstadt, University of Maribor.

Industry project (selected)

- Soil parameters for conductor analyses, 4Subsea AS
- Ossian Anchor Feasibility Assessment, SSE Renewables Offshore Wind Farm Holdings Limited
- Dogger Bank C Jacket - Geotechnical Engineering, Aibel AS
- Revolution Wind Farm (REV01) - spud and anchor analysis, Nexans
- Konsulenttjänster för bergprosjektering, prosjekt Västlänken, delprosjekt Korsvägen, Tyréns AB
- Software on in-situ data, lab data and MP design, HDEC
- Noiseless, Ørsted
- Geotechnical assessment of the capacity and stiffness of mudmat, Aker Solutions AS
- Jack-up SSA, Noble Integrator at Hugin B, Aker BP ASA
- SBJ design Avangrid's N.E. Wind, Kent
- Ginger WPS Foundations, InterMoor
- Empire wind - spud and anchor analysis, Nexans
- Baltyk Pile Assessment, Equinor

- Jack-up SSA, Martin Linge D1-H, Aker BP ASA
- ELDE Platform Mudmat, Aker Solutions AS

Teaching activity

2022	OE44030	Offshore geotechnical engineering	**Lecture**: Physical Modelling in offshore geotechnics
2021	CIE5321	Experimental Methods in Geotechnical Engineering	**Lecture**: Stress paths, Triaxial tests, Bender Element tests

Co-supervision postgraduate projects

2019-2022	PhD	Lateral response of monopile in clay	Yuen Zhang, Delft university of Technology
2022	MSc	The influence of vertical loading on the lateral behaviour of rigid monopile in clay soil	Rein de Voogd, Delft university of Technology
2022	MSc	History case and centrifuge experimental evaluation on the CPT based design models for laterally piles in sand	Daniël Veldhuijzen van Zanten, Delft university of Technology
2020	MSc	Development and Evaluation of a Sand Pluviator	Yifan Yang, Delft university of Technology

Conference/workshop organisation

2022	Geolab: Next Generation (NG) of Researchers Training Workshops November 17-18, 2022 ETH Zürich, Zürich, Switzerland
2017	The 2nd International Symposium on Coastal and Offshore Geotechnics (ISCOG) July 5-7, 2017 Zhejiang University, Hangzhou, China

Reviewing activity

Acta Geotechnica, Applied Ocean Research, Canadian Geotechnical Journal, Géotechnique Letters, International Journal Of Geomechanics, International Journal of Physical Modelling in Geotechnics, Journal of Geotechnical and Geoenvironmental Engineering, Marine Georesources & Geotechnology, Ocean Engineering, Tunnelling and Underground Space Technology, Soil Dynamics and Earthquake Engineering

Language

Mandarin (Native), English (Proficient)

Publications

Journal papers:

- Peccin da Silva A., **Wang, H.**, Zwaan R., Askarinejad, A., Pisanò, F. (2026). Cyclic lateral behaviour of monopiles in very dense sand: a centrifuge study on pile geometry and loading sequence effects. ***International Journal of Physical Modelling in Geotechnics*** (Accepted).
- Pisanò F., Kementzetzidis E., **Wang H.**, Marino M., Wu K.-W., Habib F., and Peccin da Silva A. (2026). An SCPT-based monotonic soil reaction model for monopiles in sand: calibration with centrifuge tests and validation against field data, ***Journal of Geotechnical and Geoenvironmental Engineering*** (Accepted).
- Liu H. Y., Konstadinou M., **Wang, H.**, Jostad H. P., Pisanò, F. (2026). 3D FE cyclic modelling of monopiles in sand using SANISAND-MS: calibration and validation from soil element to pile-interaction scale. ***Soil Dynamics and Earthquake Engineering***, 204, 110177.

- He, B., **Wang, H.**, Yang, S., & Nallathamby, S. (2025). Numerical evaluation of existing p-y models for the laterally loaded pile in clay. ***Marine Georesources & Geotechnology***, 1-10.
- Hu, Q., Han, F., Prezzi, M., Salgado, R., Zhao, M., **Wang, H.**, ... & Wang, L. Z. (2025). Discussion on lateral load response of large-diameter monopiles in sand. ***Géotechnique***. 75 (7): 964–967.
- **Wang, H.**, Lehane, B. M., Bransby, M. F., Pisanò, F., & Wang, L. Z. (2024). Discussion of "Numerical Modeling of the Monotonic Lateral Behavior of XL and XXL Monopiles in Kaolin". ***Journal of Geotechnical and Geoenvironmental Engineering***, 150(7), 07024007.
- **Wang, H.**, Lehane, B. M., Bransby, M. F., Wang, L. Z., Hong, Y., & Askarinejad, A. (2023). Lateral behavior of monopiles in sand under monotonic loading: Insights and a new simple design model. ***Ocean Engineering***, 277, 114334.
- Yin, X., **Wang, H.**, Pisanò, F., Gavin, K., Askarinejad, A., & Zhou, H. (2023). Deep learning-based design model for suction caissons on clay. ***Ocean Engineering***, 286, 115542.
- **Wang, H.**, Fraser Bransby, M., Lehane, B. M., Wang, L., & Hong, Y. (2023). Numerical investigation of the monotonic drained lateral behaviour of large-diameter rigid piles in medium-dense uniform sand. ***Géotechnique***, 73(8), 689-700.
- **Wang H.**, Wang L. Z., Askarinejad A., Hong Y., He B. (2022). Ultimate soil resistance of the laterally loaded pile in uniform sand. ***Canadian Geotechnical Journal***, 60(4), 587-593.
- Rui, S., Guo, Z., Wang, L., **Wang, H.**, & Zhou, W. (2022). Inclined loading capacity of caisson anchor in South China Sea carbonate sand considering the seabed soil loss. ***Ocean Engineering***, 260, 111790.
- **Wang, H.**, Lehane, B. M., Bransby, M. F., Wang, L. Z., & Hong, Y. (2022). Field and numerical study of the lateral response of rigid piles in sand. ***Acta Geotechnica***, 17(12), 5573-5584.
- **Wang, H.**, Lehane, B. M., Bransby, M. F., Askarinejad, A., Wang, L. Z., & Hong, Y. (2022). A simple rotational spring model for laterally loaded rigid piles in sand. ***Marine Structures***, 84, 103225.
- Askarinejad A., **Wang H.**, Chortis G., Gavin K. (2021). Influence of scour protection layers on the lateral response of monopile in dense sand. ***Ocean Engineering***, 244, 110377.
- **Wang H.**, Wang L. Z., Hong Y., Askarinejad A., He B., Pan H. L. (2021). Influence of Pile Diameter and Aspect Ratio on the Lateral Response of Monopiles in Sand with Different Relative Densities. ***Journal of Marine Science and Engineering***, 9(6), 618.
- **Wang H.**, Wang L. Z., Hong Y., Mašín D., Li W., He B., Pan H. L. (2021). Centrifuge testing on monotonic and cyclic lateral behavior of large-diameter slender piles in sand. ***Ocean Engineering***, 226, 108299.
- **Wang H.**, Lehane B. M., Bransby M. F., Wang L. Z., Hong Y. (2020). A simple approach for predicting the ultimate lateral capacity of a rigid pile in sand. ***Géotechnique Letters***, 10(3), 429-435.
- **Wang H.**, Wang L. Z., Hong Y., He B., Zhu R. H. (2020). Quantifying the influence of pile diameter on the load transfer curves of laterally loaded monopile in sand. ***Applied Ocean Research***, 101, 102196.
- Wang L. Z., **Wang H.**, Zhu B., Hong Y. (2018). Comparison of monotonic and cyclic lateral response between monopod and tripod bucket foundations in medium dense sand. ***Ocean Engineering***, 155, 88-105.
- He B., **Wang H.**, Hong Y., Wang L. Z., Zhao C. J., Qin X. (2016). Effect of vertical load on lateral behaviour of single pile in clay. ***Journal of Zhejiang University (Engineering Science)***, 50(7): 1221-1229.

Conference papers:

- KementzetzidisE., Cengiz C., **Wang H.**, Xie W. B., Zwaan R., Sharma A., Konstantinou M., Pisanò F., Jostad H. P., Christopoulos G., Mohapatra D., (2026). Cyclic Behaviour of Monopiles in Clay: Preliminary Insights from the MIDASclay Project. 11th International Conference on Physical Modelling in Geotechnics (ICPMG2026), Zurich, Switzerland, 2026.
- Cengiz C., Sharma A., Konstantinou M., Zwaan R., Xie W. B., Mohapatra D., **Wang H.**, Jostad H. P., KementzetzidisE., (2026). Engineering Challenges and Lessons Learnt from Centrifuge Modelling of Monopile-Clay Interaction. 11th International Conference on Physical Modelling in Geotechnics (ICPMG2026), Zurich, Switzerland, 2026.
- Zhang S. H., Wang L. Z., Zhang W. Y., Dai Z. J., Chen Z., Wang Y., **Wang H.**, Rui S. J., Xu. T., Guo Z., Li S. Z. (2026). Development of an integrated pile driving and loading. 11th International Conference on Physical Modelling in Geotechnics (ICPMG2026), Zurich, Switzerland, 2026.
- **Wang H.**, Wang Y. F., Pisanò F., Rahim A. (2025). Deeply embedded ring anchor for offshore floating structures. 5th International Symposium on Frontiers in Offshore Geotechnics (ISFOG), Nantes, France, 2025.
- **Wang H.**, Zwaan R., Peccin da Silva A., Askarinejad A., Pisanò F. (2025). Physical modelling of cyclically loaded monopiles in sand: the MIDAS centrifuge testing programme. 5th International Symposium on Frontiers in Offshore Geotechnics (ISFOG), Nantes, France, 2025.

- Kementzetzidis E., Konstantinou M., Cengiz C., Zwaan R., Sharma A., Elkadi A., Pisanò F., **Wang H.**, Jostad H. P., Christopoulos G., Mohapatra D. (2025). Centrifuge testing and numerical modelling of cyclically loaded monopiles in clay: Set up and early findings of the MIDASclay project. 5th International Symposium on Frontiers in Offshore Geotechnics (ISFOG), Nantes, France, 2025.
- Du W. H., Lehane B., Joer H., Ghanbari G., **Wang H.** (2025). Stiffness and Damping of Clay Under Cyclic Loading: An Image-Based Measurement Solution. 5th International Symposium on Frontiers in Offshore Geotechnics (ISFOG), Nantes, 2025.
- **Wang H.**, Wang Y. F., Pisanò F. (2024). Horizontal capacity of suction anchor with retrievable cap in clay for floating wind turbine. Offshore Technology Conference, Houston, US, 2024.
- **Wang H.**, He B., Nallathamby S., Yang S. L., Huynh K. (2024). Example case application of PISA method for monopile design in layered soil. GeoShanghai 2024, Shanghai, China, 2024.
- **Wang H.**, He B., Wang L. Z., Hong Y. (2024). Cyclic response of laterally loaded pile after storm loading. GeoShanghai 2024, Shanghai, China, 2024.
- **Wang H.**, Lehane B. M. (2023). Assessment of p-y approaches to estimate the lateral response of driven and vibrated piles in sand. 9th International SUT OSIG Conference "Innovative Geotechnologies for Energy Transition", London, UK, 2023.
- Kementzetzidis E., **Wang H.**, Marino M., Askarinejad A., Peccin da Silva A., Elkadi A. S., Pisanò F. (2023). Monopile-sand interaction under lateral cyclic loading: simulation of centrifuge test data using a cyclic 1D p-y model. 9th International SUT OSIG Conference "Innovative Geotechnologies for Energy Transition", London, UK, 2023.
- Yin X.L., **Wang H.**, Pisanò F., Gavin K., Askarinejad A. Zhou H.P. (2023). Application of the neural network model in predicting the three-dimensional response of suction caissons on clay. 9th International SUT OSIG Conference "Innovative Geotechnologies for Energy Transition", London, UK, 2023.
- **Wang H.**, Veldhuijzen van Zanten D., de Lange D., Pisanò F., Gavin K., Askarinejad A. (2022). Centrifuge study on the CPT based *p-y* models for the monopiles. 5th International Symposium on Cone Penetration Testing, Bologna, 2022.
- Wang L.Z., Lai Y. Q., Hong Y., **Wang H.** (2021) A unified lateral soil reaction model for monopiles in soft clay and sand considering various length-to-diameter (*L/D*) ratios. Proceedings of the 20th International Conference on Soil Mechanics and Geotechnical Engineering, Sydney 2021.
- **Wang H.**, Wang L.Z., Hong Y. He B., Lai Y. Q. (2021). Deflection recovery of large diameter monopile under cyclic loading for offshore wind turbines. Proceedings of the 20th International Conference on Soil Mechanics and Geotechnical Engineering, Sydney 2021.
- Pisanò F., Askarinejad A., Gavin K., **Wang H.**, Maghsoodi S., Segeren M., Elkadi A., de Lange D., Konstantinou M. (2021) MIDAS: Monopile Improved Design through Advanced cyclic Soil modelling. Proceedings of the 20th International Conference on Soil Mechanics and Geotechnical Engineering, Sydney 2021.
- **Wang H.**, Lehane B. M., Bransby M. F., Wang L. Z., Hong Y. (2020). An Investigation Of the contribution of the pile base to the lateral response of monopiles under static loading. 4th International Symposium on Frontiers in Offshore Geotechnics (ISFOG), Texas, 2022.
- **Wang H.**, Wang L.Z., Hong Y. (2018). Centrifuge Study on the Lateral Loaded Response of the Monopod and Tripod Bucket Foundations in Sand. In Vietnam Symposium on Advances in Offshore Engineering (pp. 428-433). Springer, Singapore.

Presentations:

- **Wang H.** Lateral behaviour of monopiles under monotonic loading in drained sand: insights and a simple design model. OSIGp Prize 2023 Awarding Ceremony, Perth, Australia, January 23, 2024.
- **Wang H.** Application of physical modelling in offshore geotechnics: some examples. (Keynote) Presented at: Geolab: Next Generation (NG) of Researchers Training Workshops, Oslo, Norway, November 30, 2023.
- **Wang H.** Application of physical modelling in offshore geotechnics: some examples. (Keynote) Presented at: Geolab: Next Generation (NG) of Researchers Training Workshops, Zürich, Switzerland, November 17-18, 2022.
- Askarinejad A., **Wang H.** Application of physical modelling in offshore geotechnics: some examples. (Keynote) Presented at: Geolab: Next Generation (NG) of Researchers Training Workshops, Madrid, Spain, December 2-3, 2021.
- **Wang H.** Lateral response of large-diameter slender piles in sand supporting offshore wind turbines. Presented at: Geo-talks of Geo-section at TU Delft, Delft, Netherlands, December 9, 2020.
- **Wang H.**, Lehane B. M., Bransby M. F. Lateral response of monopiles under static loading. Presented at: 2019 COFS/CEME Research Workshop, Perth, Australia, November 3-4, 2019.

- **Wang H.**, Hong Y., Wang L. Z. Comparison of monotonic and cyclic lateral response between monopod and tripod bucket foundations in medium dense sand. Presented at: International Symposium on Energy Geotechnics (SEG-2018), Lausanne, Switzerland, September 25-28, 2018.
- **Wang H.**, Hong Y., Wang L. Z. Numerical investigation on the failure mechanism of the tripod bucket foundation under lateral loading in medium dense sand. Presented at: The 2nd International Symposium on Coastal and Offshore Geotechnics, Hangzhou, China, July 5-7, 2017.